Agentic AI-Based Multi-Agent Recommender System for Higher Education Decision Support

Samuel Shine
Department of Computer Science (MCA)
CHRIST (Deemed to be University)
Bengaluru, India
samuelshine112003@gmail.com

***Abstract*—Students choosing which university and program to apply to must jointly reason over academic fit, career goals, financial constraints, location preferences, and employability outcomes, typically before they have generated any interaction history a recommender can learn from. Classical educational recommender systems address this pre-admission, cross-institution decision using collaborative, content-based, or knowledge-based filtering, but remain static, weakly explainable, and brittle under cold start. Recent agentic, LLM-powered multi-agent recommenders address adaptivity, planning, memory, and explainability, but published 2025–2026 systems apply these capabilities to *already-enrolled* students inside an *already-chosen* program (academic advising, learning-pathway sequencing, in-curriculum course selection) rather than to the pre-admission choice itself. This paper identifies that intersection as an open gap, proposes a five-agent architecture (profiling and constraint elicitation, retrieval, multi-objective recommendation and planning, explanation, and memory-reflection) targeting pre-admission cross-institution decision support, and lays out a research design – research questions, hypotheses, and an evaluation protocol against classical baselines on standard and agentic-specific metrics – for the prototype currently under development. [TODO: Replace this paragraph with a results-bearing abstract once the prototype and evaluation in Sections IV–V are complete.]**

***Index Terms*—agentic AI, multi-agent systems, recommender systems, higher education, large language models, explainable recommendation, decision support**

I. Introduction

Choosing a university and program is a high-stakes, low-frequency decision made under incomplete information. Prospective students must reconcile academic interests, career goals, skills, financial constraints, location preferences, and employability outcomes, usually without any prior interaction history a recommender system could exploit – the canonical cold-start setting. A substantial body of work addresses this decision with collaborative filtering, content-based filtering, knowledge-based recommendation, and hybrid or machine-learning classification approaches [1]–[10]. These systems established strong evaluation methodology, real datasets, and education-specific features, but a 2024 systematic review of the space documents recurring limitations: static recommendation pipelines, limited personalization, minimal conversational interaction, weak adaptation over time, and weak explainability [11].

Separately, the emergence of large language model (LLM)-powered agents has produced a new paradigm for recommendation: systems that plan, retrieve, reflect, and maintain long-term memory across a conversation rather than scoring a fixed candidate set once [12]–[14]. Agentic recommenders of this kind have been demonstrated in e-commerce and general content domains [15]–[18], and, more recently, in education – but specifically for students who are *already enrolled* in an *already-chosen* program: academic advising Q&A [19], in-program learning-pathway sequencing [20], and in-curriculum course selection [21]. None of these systems address the earlier, arguably higher-stakes decision of *which* institution and program to choose in the first place.

This paper's central claim is that this earlier decision – pre-admission, cross-institution university and program selection – is a distinct decision scope from the in-program advising/pathway/course-selection problems that 2025–2026 agentic education systems solve, and that it remains addressed only by pre-agentic methods. Section II substantiates this claim against the closest prior work in both the classical educational recommender literature and the agentic recommender literature. Section III proposes a five-agent architecture for this gap. Section IV and Section V lay out the research questions, hypotheses, and evaluation protocol guiding the prototype and experiments currently under development.

This paper's contributions are:

1) An agentic multi-agent LLM architecture targeting pre-admission, cross-institution higher-education decision support – a decision scope distinct from the in-program advising/pathway/course systems in recent agentic-education work (Section II).
2) An explicit multi-objective reasoning mechanism integrating career goals, financial constraints, location, and employability into agent planning, to be evaluated against classical educational-RS baselines on both standard information-retrieval metrics and agentic-specific metrics (explanation quality, adaptivity, cold-start robustness), answering the "lack of educational evaluation frameworks" gap named explicitly in [12].
3) **[TODO: A working prototype and evaluation protocol, released as an open-source artifact that third parties can run (repository URL to be added) – finalize this claim once Sections IV–V are executed and the repository is public.]**

II. Related Work

A. Classical educational recommenders for pre-admission decisions

HRSPCA [1] and StudieMe [2] are representative of the pre-admission, cross-institution decision scope this paper targets: both recommend which college/university a prospective student should consider, using hybrid collaborative filtering with knowledge-based rules [1] or content-based TF-IDF similarity [2]. Somyürek and Aksoy [7] extend this to joint university-and-major selection with a multidimensional recommender, the closest classical match to the decision scope adopted here. Elahi et al. [9] develop a collaborative-filtering university recommender evaluated in two phases – offline cross-validation followed by an online user study measuring accuracy, diversity, perceived personalization, satisfaction, and novelty – an evaluation protocol this paper adopts as a template (Section V). Lahoud et al. [8] compare several such systems for major/career guidance and report fragmentation across methods rather than a unified adaptive framework; El-Keiey et al. [22] address the adjacent career-path decision with feature-selected machine-learning classification over a public 20,000-record student-features dataset, and Hill et al. [10] recommend academic programs to prospective students via BERTopic interest mining – both from 2025, and both still single-model, non-agentic pipelines. None of these systems reason via autonomous planning agents, maintain memory across sessions, or generate grounded natural-language explanations; Kamal et al.'s systematic review [11] confirms these as recurring, unresolved limitations across the sub-field. By contrast, [23] and [24] address a related but distinct *post-enrollment* scope (in-program course selection), which this paper does not target.

B. Agentic and multi-agent recommenders (general domain)

Portugal et al. [15] propose an agentic multi-agent recommender framework whose title closely resembles this paper's working title, making it the closest general-domain precedent; it targets e-commerce recommendation and does not incorporate educational features, datasets, or evaluation metrics. Yousefi Maragheh et al. [16] survey multi-agentic design patterns for industrial recommendation, and Narasimhan and Narasimhan [17] introduce a verifiable benchmark for agentic recommenders, evaluated on general-domain conversational tasks across five model families – no education-specific agentic-recommendation benchmark yet exists. Peng et al.'s survey [12] categorizes agentic recommendation into recommender-oriented, interaction-oriented, and simulation-oriented paradigms and explicitly identifies limited higher-education application as an open direction, which this paper addresses directly.

C. Agentic multi-agent systems in education

Multi-agent educational recommenders predate LLMs: Marivate et al. [25] deployed cooperative collaborative-filtering agents for professional course selection in 2008, and Viswanathan [26] surveys the pre-LLM adaptive multi-agent e-learning line. The novelty claimed here therefore rests on the combination of *agentic LLM* capabilities (planning, memory, reflection, grounded explanation) with the pre-admission decision scope – not on multi-agent structure alone. Three 2025–2026 systems are the decisive comparison set, since each combines multi-agent LLM methods with an educational target. AdvisingWise [19] is a human-in-the-loop multi-agent system that answers academic-advising questions for *already-enrolled* graduate students within a fixed program, with all responses validated by a human advisor before delivery – it does not recommend which program to choose, and its human-in-loop response drafting differs from autonomous multi-agent planning. Chaturvedi and Gunawardena [20] deploy an 18-agent, four-tier hierarchical architecture for adaptive in-program learning-pathway recommendation at production scale (6,000+ users), but likewise address only *post-enrollment* curriculum adaptation. Zhong et al. [21] propose a three-agent architecture (learner profiling, knowledge retrieval, recommendation generation) coupled with a course knowledge graph for *in-curriculum* course recommendation – structurally the closest precedent to the architecture proposed in Section III, but scoped to course-level choice for enrolled students rather than institution/program-level choice for prospective students, and without multi-objective reasoning over career, financial, location, and employability constraints.

TABLE I
Positioning against the closest prior work

System	Target user	Decision scope	Method
[1], [2], [7]	Prospective	Pre-admission, cross-institution	CF / CB / KB (pre-agentic)
[19]	Enrolled	In-program advising	Multi-agent, human-in-loop
[20]	Enrolled	In-program pathway	Multi-agent (18+ agents)
[21]	Enrolled	In-curriculum course	Multi-agent + KG (3 agents)
This work	**Prospective**	**Pre-admission, cross-institution**	**Multi-agent LLM (5 agents)**

D. Positioning

Table I summarizes the resulting gap: no system in the surveyed literature combines an agentic multi-agent architecture with the pre-admission, cross-institution decision scope while reasoning jointly over the constraint set a prospective student actually faces.

III. Proposed Architecture

The initial brainstorm for this project proposed seven agents (profiling, goal planning, knowledge retrieval, recommendation, reflection, explanation, memory). Two of the closest comparison systems bound a reasonable agent count: Zhong

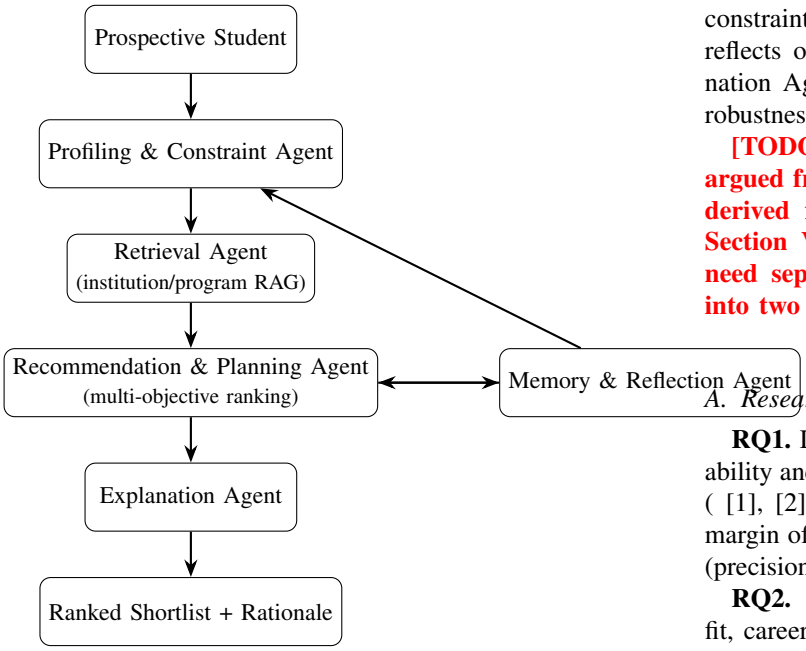


Fig. 1. Proposed five-agent architecture. The Memory & Reflection Agent persists the student's constraint object across sessions and reviews the Recommendation & Planning Agent's shortlist before it reaches the Explanation Agent.

et al. [21] report accuracy, recall, and F1 gains with only three agents, while Chaturvedi and Gunawardena [20] use 18+ agents but only at production deployment scale with a dedicated engineering team – not reproducible within a single-author prototype. This paper accordingly merges the original seven agents into five, keeping explanation and memory-reflection as separate, visible components because they correspond directly to this paper's headline claims (explainability, cold-start robustness via memory), while folding goal-planning's constraint logic into profiling and recommendation.

Profiling & Constraint Agent. Elicits academic background, career goals, and financial, location, and employability-outcome constraints through structured dialogue; outputs a structured student-constraint object consumed by downstream agents.

Retrieval Agent. Performs retrieval-augmented generation over institution and program data (admissions criteria, cost, location, outcome statistics), following the retrieval-plus-generation pattern established in [18] but against an education-specific knowledge base.

Recommendation & Planning Agent. Ranks retrieved candidates against the student-constraint object under multiple, potentially conflicting objectives (academic fit, financial feasibility, location, employability outcome) – the mechanism RQ2 (Section IV) evaluates directly.

Explanation Agent. Generates a per-recommendation natural-language rationale grounded in the evidence the Retrieval Agent surfaced, addressing the explainability limitation [11] documents in classical systems.

Memory & Reflection Agent. Persists the student-constraint object and prior shortlists across sessions, and reflects on the current shortlist before it reaches the Explanation Agent – the mechanism RQ3 evaluates for cold-start robustness.

[TODO: This merge (7→5 agents) is a judgment call, argued from buildability within a one-month timeline, not derived from an ablation. Revisit after the ablation in Section V (H3) – if memory and reflection turn out to need separable evaluation, they may need to split back into two agents.]

IV. Research Design

A. Research questions

RQ1. Does the architecture in Section III improve explainability and adaptivity over classical educational recommenders ([1], [2], [7]) while remaining within a 5% non-inferiority margin of their accuracy on standard recommendation metrics (precision@k, recall@k, NDCG)?

RQ2. How should multi-objective constraints (academic fit, career goals, financial constraints, location, employability outcomes) be represented and reasoned over by specialized agents, compared to a single monolithic LLM prompt or a classical hybrid scoring function?

RQ3. Do the Memory & Reflection Agent's mechanisms measurably reduce cold-start degradation for prospective students with no prior interaction history, compared to a stateless agent pipeline?

B. Hypotheses

H1. The proposed architecture achieves higher user-perceived explanation quality than a classical hybrid baseline re-implemented from [1], [2], while remaining within a pre-registered 5% *relative* non-inferiority margin on NDCG@k (headline metric; precision@k and recall@k reported as secondary), on a held-out admissions/preference dataset.

H2. Explicit multi-objective agent reasoning improves recommendation relevance for students with conflicting constraints (e.g., financial limits vs. top academic fit) relative to a single-objective ranking baseline.

H3. The Memory & Reflection Agent mitigates cold-start degradation more effectively than a stateless agent pipeline, measured across simulated first-session vs. later-session interactions.

C. Planned methodology

Evaluation uses a dual-dataset design (decided 2026-07-16, plan-mode dialogue; see `chapter_plan.md`): (i) the university-rating dataset of [9] (1,515 ratings, 80 users, 551 universities, plus metadata for 12,003 crawled universities) as the primary accuracy benchmark for H1 – real preference data at exactly this paper's decision scope (requested from the authors; request pending); and (ii) a synthetic constraint-preference dataset, seeded from the public 20,000-record student-features corpus of [22] and public program/admissions data following [7], as the testbed for H2 and H3 – conflicting-constraint and multi-session cold-start conditions cannot be

expressed in rating-only data, and the scarcity of public educational datasets with such structure is documented in [27]. **[TODO: If the [9] data is not obtained in time, the synthetic dataset carries H1 as well – in that case state the self-constructed-benchmark limitation explicitly in Section VII.]**

[TODO: Baselines (fidelity decision 2026-07-16: faithful re-implementation "as it was", not library proxies): (a) a re-implemented hybrid CF+KB baseline following [1]; (b) a re-implemented content-based baseline following [2]; (c) a single-prompt LLM baseline (no multi-agent decomposition) to isolate the effect of the architecture itself, following the ablation logic in [21]. Where the original papers underspecify a detail, a best-faith reconstruction choice is made and logged in an appendix reconstruction table. Prototype and experiments run on a local open-weights Gemma model via Ollama (tag `gemma4:e4b`; `gemma4:12b` available for an optional model-scale ablation); the released artifact is provider-agnostic (bring-your-own API key or local model).]

V. Evaluation Plan

Metrics. The headline metric is NDCG@5 with graded relevance, under the pre-registered 5% relative non-inferiority margin against the strongest classical baseline (H1); precision@5 and recall@5 are reported as secondary metrics, following [28]. All systems are evaluated on the same held-out split of 100 students; the remaining 200 students serve as the historical pool for the HRSPCA predictor stage.

Explanation quality is scored by LLM-as-judge on a three-dimension rubric (clarity, groundedness in the provided program evidence, usefulness; 1–5 each), following the protocol direction of [19]. To avoid self-preference bias – same-family judges systematically inflate scores – the judge is drawn from a different model family than the experiment model (primary judge: Qwen-family 14B-class; a Mistral-family second judge provides inter-judge agreement), with system identity blinded and absolute rather than pairwise scoring to avoid position bias. LLM-as-judge validity is treated as a limitation in Section VII.

Conditions. (i) the full five-agent pipeline; (ii) a *no-memory* ablation (H3); (iii) a *single-objective* ablation collapsing the Recommendation & Planning Agent's reasoning to academic fit only (H2); (iv) a *single-prompt* LLM baseline replacing the five-agent decomposition, following the ablation logic in [21]; and (v) the two re-implemented classical baselines. H1 comparisons use the full-information state (complete profile, mirroring the filled application form classical systems assume); H3 uses the three-session progressive-disclosure structure with simulated-user feedback reacting to what each system actually recommended in the prior session.

VI. Results

[TODO: This section reports completed runs only; agentic-condition rows are pending the evaluation batches currently executing and MUST be filled from run records, never estimated.]

TABLE II
Accuracy on the held-out evaluation split ($n = 100$, full information). Baseline rows are completed runs; remaining rows pending.

System	NDCG@5	P@5	R@5
HRSPCA (re-impl. [1])	0.793	0.796	0.061
StudieMe (re-impl. [2])	0.673	0.760	0.113
Single-prompt LLM baseline	**[TODO: pend.]**	**[TODO: pend.]**	**[TODO: pend.]**
Five-agent (proposed)	**[TODO: pend.]**	**[TODO: pend.]**	**[TODO: pend.]**

A. Top-k accuracy (H1)

Table II reports accuracy on the held-out 100-student evaluation split at the full-information state. The re-implemented classical baselines (Section IV-C, reconstruction table in the appendix) set the non-inferiority bar: the strongest baseline, HRSPCA, reaches NDCG@5 $= 0.793$, so under the pre-registered 5% relative margin the proposed system must reach NDCG@5 ≥ 0.753 for H1's accuracy arm to hold.

B. Explanation quality (H1), multi-objective conflict (H2), and cold start (H3)

[TODO: Explanation-quality rubric scores (both judges + agreement), the single-objective ablation on the 162-student conflict subset (H2), and per-session accuracy trajectories for full vs. no-memory conditions (H3) are pending their respective batches. Report exactly what the run records and judge outputs contain; where a hypothesis fails, report the failure (pre-committed in the evaluation design).]

VII. Discussion and Limitations

The gap argument in Section II was verified against full-text readings of the decisive prior work: all three agentic-education systems [19]–[21] were confirmed to address only already-enrolled students (advising, learning pathways, course selection), and the two re-implemented classical baselines [1], [2] were confirmed to contain no agentic, dialogue, or memory elements. Peripheral Cluster A citations remain verified at the abstract level only. The 7-to-5 agent merge (Section III) is a buildability judgment, not an empirical result, and may not survive the ablation in H3.

Beyond the gap-verification scope, this work carries the following limitations, stated before results are in so that none can be read as post-hoc rationalization.

Oracle circularity. Ground-truth relevance in the synthetic dataset is a published utility function fixed before any system was evaluated, not human judgment. A system that reverse-engineered the formula would score perfectly. Mitigations: the oracle is documented in the released dataset card, identical for all compared systems, and H1's primary benchmark remains the real-preference dataset of [9] if access is granted; explanation quality is judged independently of the oracle.

Synthetic evaluation data. The program catalog is fictional (real cities, invented institutions, calibrated but invented economics), student constraints are attached by documented sampling rules, and the three-session structure is scripted rather than observed user behavior, with simulated-user feedback derived from the oracle. Career coverage is IT-only, inherited from the seed corpus of [22]; generalization beyond IT-bound students is untested. The scarcity of public educational datasets motivating this choice is itself documented [27].

Baseline reconstruction fidelity. Both classical baselines were re-implemented from their papers' full texts, with every underspecified detail resolved by a logged best-faith choice (appendix reconstruction table). Two original components could not be expressed in the evaluation catalog (HRSPCA's college-capacity rule; StudieMe's faculty-matching stage); both omissions remove hard filters and therefore favor the baselines, not the proposed system.

LLM-as-judge validity. Explanation-quality scores come from LLM judges, not human raters. Cross-family judging, blinding, absolute scoring, and two-judge agreement reduce – but do not eliminate – known judge biases; no human-subject validation is performed within this paper's timeline, and the judge scores should be read accordingly.

Single experiment model and scale. All agentic conditions run on one local open-weights model (`gemma4:e4b`); conclusions about the architecture are conditioned on this model class. The evaluation split is $n = 100$ students against a 200-program catalog – larger than the closest classical baselines' user studies [9], [10] but small in absolute terms.

VIII. Conclusion and Future Work

This paper identified pre-admission, cross-institution university and program selection as a decision scope left unaddressed by both classical educational recommenders (pre-agentic methods) and 2025–2026 agentic education systems (in-program scope only), proposed a five-agent architecture for this gap, and laid out the research questions, hypotheses, and evaluation protocol that will be executed next. **[TODO: Replace this paragraph with a findings-bearing conclusion once Sections IV–V are complete.]**

Acknowledgment

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Data Availability Statement

The synthetic evaluation dataset (students, program catalog, graded relevance, session structure), its generator, and its dataset card documenting provenance, the ground-truth oracle, and known limitations are released with the paper's open-source artifact. The public seed corpus of [22] is fetched from its original GPL-3.0 repository at generation time and is not redistributed. **[TODO: Add the artifact repository URL; if the [9] dataset is obtained and used for H1, state its on-request availability and usage conditions here.]**

Author Contributions

Single-author work; CRediT taxonomy roles apply in full to the sole author.

Conflict of Interest Statement

The author declares no conflict of interest.

AI-Assisted Writing Disclosure

[TODO: This draft was produced with AI assistance (literature search, drafting, LaTeX formatting) under author supervision. Replace with the venue-specific disclosure statement required by the target IEEE conference before submission – wording differs by venue.]

References

[1] A. H. M. Ragab, A. F. S. Mashat, and A. M. Khedra, "Hrspca: Hybrid recommender system for predicting college admission," in *2012 12th International Conference on Intelligent Systems Design and Applications (ISDA)*, 2012, pp. 107–113.

[2] V. Sharma, T. Trehan, R. Chanana, and S. Dawn, "Studieme: College recommendation system," in *2019 3rd International Conference on Recent Developments in Control, Automation & Power Engineering (RDCAPE)*, 2019, pp. 227–232.

[3] S. J. Fong and R. P. Biuk-Aghai, "An automated university admission recommender system for secondary school students." [Online]. Available: https://api.semanticscholar.org/CorpusID:49586332

[4] J. R, J. Christy, A. D, A. V, and A. J, "Smart college admission recommendation system," 08 2025, pp. 666–670.

[5] S. Kanoje, D. Mukhopadhyay, and S. Girase, "User profiling for university recommender system using automatic information retrieval," *Procedia Computer Science*, vol. 78, pp. 5–12, 12 2016.

[6] M. Mokarrama, S. Khatun, and M. Arefin, "A content-based recommender system for choosing universities," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 28, pp. 2128 – 2142, 07 2020.

[7] S. Somyürek and N. Aksoy, "Navigating academia: Designing and evaluating a multidimensional recommendation system for university and major selection," *Psychology in the Schools*, vol. 61, pp. 3748–3769, 06 2024.

[8] C. Lahoud, S. Moussa, C. Obeid, H. Khoury, and P.-A. Champin, "A comparative analysis of different recommender systems for university major and career domain guidance," *Education and Information Technologies*, vol. 28, 12 2022.

[9] M. Elahi, A. Starke, N. El Ioini, A. A. Lambrix, and C. Trattner, "Developing and evaluating a university recommender system," *Frontiers in Artificial Intelligence*, vol. 4, p. 796268, 2022.

[10] A. Hill, K. Goo, and P. Agarwal, "Recommending the right academic programs: An interest mining approach using bertopic," 2025. [Online]. Available: https://arxiv.org/abs/2501.06581

[11] N. Kamal, F. Sarker, A. Rahman, S. Hossain, and K. A. Mamun, "Recommender system in academic choices of higher education: A systematic review," *IEEE Access*, vol. 12, pp. 35 475–35 501, 2024.

[12] Q. Peng, H. Liu, H. Huang, Q. Yang, and M. Shao, "A survey on llm-powered agents for recommender systems," 2025. [Online]. Available: https://arxiv.org/abs/2502.10050

[13] D. B. Acharya, K. Kuppan, and B. Divya, "Agentic ai: Autonomous intelligence for complex goals—a comprehensive survey," *IEEE Access*, vol. 13, pp. 18 912–18 936, 2025.

[14] H. Derouiche, Z. Brahmi, and H. Mazeni, "Agentic ai frameworks: Architectures, protocols, and design challenges," 2025. [Online]. Available: https://arxiv.org/abs/2508.10146

[15] I. D. S. Portugal, P. Alencar, and D. Cowan, "An agentic ai-based multi-agent framework for recommender systems," in *2024 IEEE International Conference on Big Data (BigData)*, 2024, pp. 5375–5382.

[16] R. Yousefi Maragheh, Y. Deldjoo, C. Wang, J. Cho, and D. Cheng, "Multi-agentic recommender systems: Foundations, design patterns, and e-commerce applications — an industrial tutorial," in *Proceedings of the Nineteenth ACM Conference on Recommender Systems*, ser. RecSys '25. New York, NY, USA: Association for Computing Machinery, 2025, p. 1427–1429. [Online]. Available: https://doi.org/10.1145/3705328.3748008

[17] B. S. Narasimhan and K. R. Narasimhan, "τ-rec: A verifiable benchmark for agentic recommender systems," 2026. [Online]. Available: https://arxiv.org/abs/2606.10156

[18] F. S. Alamri, A. Rehman, B. Alghofaily, A. Ahmed, and K. Saleem, "Leveraging rag with transformer for context-based personalized recommendations," *IEEE Access*, vol. 13, pp. 97 472–97 486, 2025.

[19] W. Jiang, S. Wang, H. Eltigani, R. Haroon, A. Bin Faisal, and F. Dogar, "Advisingwise: Supporting academic advising in higher education settings through a human-in-the-loop multi-agent framework," 2025. [Online]. Available: https://arxiv.org/abs/2511.05706

[20] N. Chaturvedi and A. Gunawardena, "Agentic orchestration for adaptive educational recommendations: A multi-agent llm framework for personalized learning pathways," in *Proceedings of the Nineteenth ACM International Conference on Web Search and Data Mining (WSDM Companion '26)*, 2026.

[21] Z. Zhong, C. Shao, Y. Hong, and Q. Li, "Online course recommendation based on multi-agent systems and knowledge graphs," in *Proceedings of the 2025 2nd International Symposium on Artificial Intelligence for Education*, 2025.

[22] S. El-Keiey, D. ElMenshawy, and E. Hassanein, "Career recommendation based on feature selection for undergraduate students using machine learning techniques," *International Journal of Advanced Computer Science and Applications*, vol. 16, no. 3, pp. 232–238, 2025.

[23] Z. A. Pardos and W. Jiang, "Designing for serendipity in a university course recommendation system," in *Proceedings of the Tenth International Conference on Learning Analytics & Knowledge*, ser. LAK '20. New York, NY, USA: Association for Computing Machinery, 2020, p. 350–359. [Online]. Available: https://doi.org/10.1145/3375462.3375524

[24] A. Parameswaran, P. Venetis, and H. Garcia-Molina, "Recommendation systems with complex constraints: A course recommendation perspective," *ACM Trans. Inf. Syst.*, vol. 29, no. 4, Dec. 2011. [Online]. Available: https://doi.org/10.1145/2037661.2037665

[25] V. N. Marivate, G. Ssali, and T. Marwala, "An intelligent multi-agent recommender system for human capacity building," in *MELECON 2008 - The 14th IEEE Mediterranean Electrotechnical Conference*, 2008, pp. 909–915.

[26] N. Viswanathan, "Adaptive multi-agent e-learning recommender systems," 2020. [Online]. Available: https://arxiv.org/abs/2012.09342

[27] F. L. da Silva, B. K. Slodkowski, K. K. A. da Silva, and S. C. Cazella, "A systematic literature review on educational recommender systems for teaching and learning: research trends, limitations and opportunities," *Education and Information Technologies*, vol. 28, no. 3, pp. 3289–3328, 2023.

[28] F. Isinkaye, Y. Folajimi, and B. Ojokoh, "Recommendation systems: Principles, methods and evaluation," *Egyptian Informatics Journal*, vol. 16, no. 3, pp. 261–273, 2015. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1110866515000341